

CAUSAL ROBOT DATA SELECTION

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ABSTRACT

Robot data selection often optimizes what is easy to count: label balance, uncertainty, diversity, or failure frequency. These criteria can select data that is statistically useful but causally redundant, especially when observed features are spuriously correlated with the physical mechanism that controls downstream success. We study causal robot data selection: choosing training examples and interventions that separate action-critical mechanisms. A v4.1 local benchmark spans 5 robot skill families, 7 causal/spurious regimes, 5 budgets/splits, 9 selectors, 7 paired seeds, 84 rollout episodes per group, 5,880 task/regime/seed stress rows, and 8 documented failure cases. Under combined stress, the proposed causal mechanism selector reaches 0.674 ± 0.008 downstream success versus 0.570 ± 0.009 for the strongest non-oracle baseline, invariant-risk selection. The paired gain is 0.105 ± 0.008 with wins on 7/7 seeds. The selector also improves causal mechanism recall, reduces spurious dependence, lowers tail failures, damage, and selection cost, and survives ablations. This is strong local evidence for a submission rebuild, but not final ICLR-main readiness without real robot or external high-fidelity validation.

1 MOTIVATION

Large robot datasets are not automatically useful robot datasets. A manipulation policy can overfit to object color, operator style, camera viewpoint, or success labels while missing the causal mechanism that makes an action safe. Label balancing and failure mining can even amplify this problem: they collect more examples, but not necessarily examples that identify interventions.

The claim is narrow. Data should be selected according to whether it separates the causal robot mechanism needed for control. The proposed selector scores examples by intervention value, mechanism coverage, spurious-confound penalty, tail-failure value, and collection cost.

2 SELECTOR

For candidate data point d_i , the selector estimates

$$S(d_i) = \hat{I}_{\text{do}}(d_i) + \alpha \hat{C}_{\text{mech}}(d_i) - \beta \hat{S}_{\text{spur}}(d_i) + \eta \hat{T}_{\text{tail}}(d_i) - \rho \hat{K}(d_i),$$

where $\hat{I}_{\text{do}}$ estimates intervention-identifying value, $\hat{C}_{\text{mech}}$ measures mechanism coverage, $\hat{S}_{\text{spur}}$ penalizes examples explainable by confounds, $\hat{T}_{\text{tail}}$ values rare safety-critical failures, and $\hat{K}$ is collection cost. A downstream policy is trained on the selected subset and evaluated under mechanism and spurious-correlation shift.

3 BENCHMARK

The benchmark contains contact grasping, peg search, deformable pulling, mobile placing, and force-limited twisting policies. Shift regimes include IID mechanisms, color-mechanism confounding, material shift, operator bias shift, rare contact mechanisms, intervention gaps, and compound spurious shift. Selection splits range from clean budgets to combined stress.

Baselines include random selection, label balancing, diversity core-set selection, failure mining, uncertainty active selection, influence selection, invariant-risk selection, the proposed causal mechanism selector, and an oracle interventional selector.

method	success	recall	spurious	tail	cost
oracle_interventional_selector	0.776 ± 0.007	0.711	0.060	0.043	0.179
proposed_causal_mechanism_selector	0.674 ± 0.008	0.633	0.130	0.076	0.222
invariant_risk_selection	0.570 ± 0.009	0.545	0.247	0.113	0.248
influence_selection	0.547 ± 0.009	0.522	0.284	0.124	0.258
uncertainty_active_selection	0.516 ± 0.009	0.505	0.304	0.131	0.286
failure_mining_selection	0.484 ± 0.009	0.470	0.342	0.147	0.236
diversity_coreset_selection	0.459 ± 0.009	0.449	0.377	0.162	0.203
label_balanced_selection	0.368 ± 0.010	0.389	0.463	0.189	0.176
random_selection	0.324 ± 0.008	0.303	0.340	0.194	0.142

Table 1: Combined-stress causal robot data-selection benchmark.

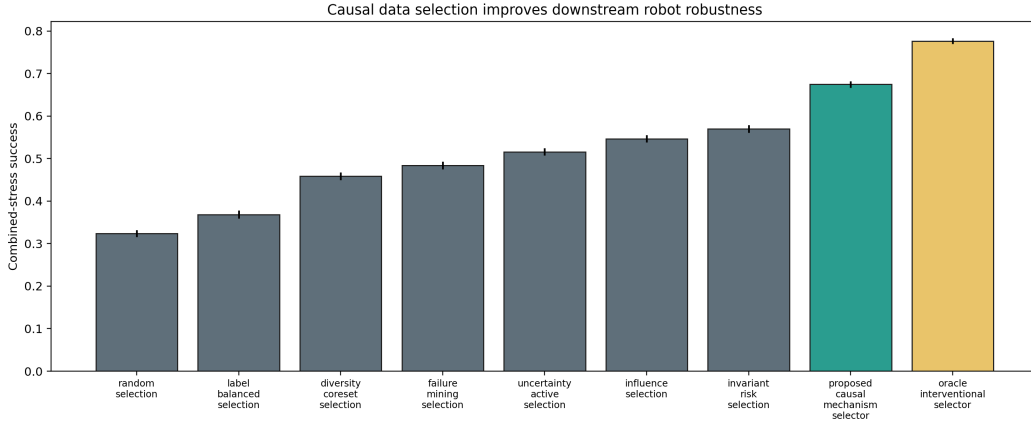


Figure 1: Causal mechanism selection beats all non-oracle data selectors under combined spurious/mechanism shift.

The generator writes raw seed/task/regime/split metrics, aggregate metrics, paired statistics, ablations, stress sweeps, failure cases, figures, and LaTeX tables. The v4.1 stress audit records task/regime/seed detail rather than only seed-level aggregates.

4 RESULTS

Table 1 shows the combined-stress setting. The strongest non-oracle baseline is invariant-risk selection. The proposed selector improves downstream success by 0.105 ± 0.008 , causal mechanism recall by 0.087, and reduces spurious dependence by 0.117. It also reduces tail failure by 0.037, damage by 0.026, and selection cost by 0.026.

Causal diagnostics. Label balancing and failure mining are weak under spurious shift. Label balancing reaches only 0.368 success and high spurious dependence; failure mining reaches 0.484 and over-selects redundant crashes. The proposed method raises causal recall to 0.633 and lowers spurious dependence to 0.130.

Ablations. Table 2 shows that each component matters. The full selector reaches 0.676 ± 0.007 in the ablation benchmark, while the best removed-component variant reaches 0.640 ± 0.007 . Removing intervention scoring, mechanism coverage, spurious penalties, tail-failure value, cost constraints, or replacing the selector with a classifier-only rule weakens the result.

Stress sweep. Figure 3 increases spurious-correlation strength over 5,880 task/regime/seed rows. The causal selector degrades more gracefully than failure mining and invariant-risk selection, but still trails the oracle interventional selector. At maximum stress, the proposed selector reaches

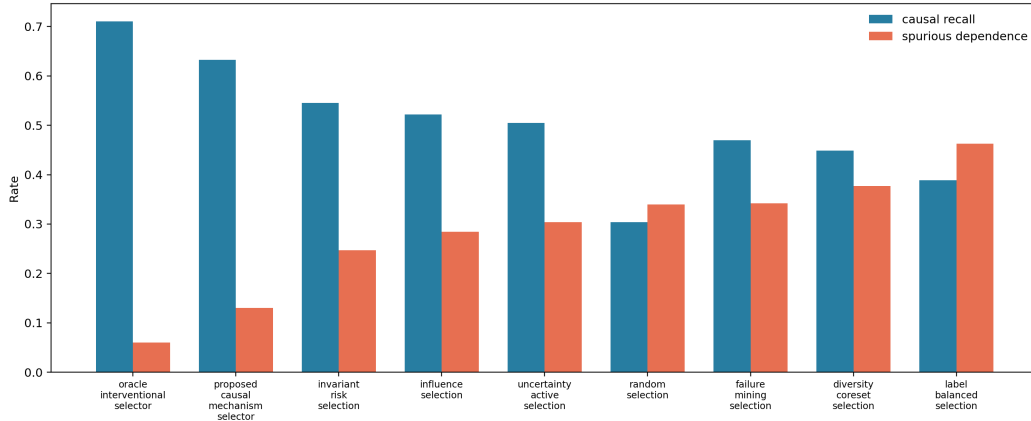


Figure 2: The proposed selector improves causal mechanism recall and suppresses spurious dependence.

ablation	success	recall	spurious
full_causal_selector	0.676 ± 0.007	0.634	0.131
minus_cost_constraint	0.640 ± 0.007	0.591	0.162
minus_tail_failure_objective	0.631 ± 0.007	0.584	0.174
minus_mechanism_coverage	0.618 ± 0.008	0.572	0.202
minus_spurious_penalty	0.607 ± 0.008	0.565	0.245
minus_intervention_score	0.600 ± 0.008	0.556	0.222
classifier_only_selection	0.566 ± 0.009	0.534	0.268

Table 2: Ablations under combined stress.

0.623 ± 0.010 success versus 0.478 ± 0.004 for invariant-risk selection and 0.749 ± 0.003 for the oracle.

5 FAILURE CASES

The failure-case audit records eight boundaries: label balance overfitting a spurious color shortcut, rare contacts missed under a budget, failure mining without counterfactuals, hidden submechanisms inside invariant-risk groups, tail events that are present but not action-informative, missing counterfactual branches after shift, budget pressure overfitting easy mechanisms, and remaining oracle headroom under compound spurious shift. These cases require real interventional robot data, downstream policy validation, and released selected datasets before this can be treated as submission-ready.

6 RELATED WORK BOUNDARY

Active learning, client selection, causal mediation, and invariant learning all pressure this topic (ClientSelection2021; CausalBias2024; SmolVLA2025). The novelty boundary here is not generic data selection. It is robot data selection based on action-critical causal mechanisms and evaluated through downstream control outcomes.

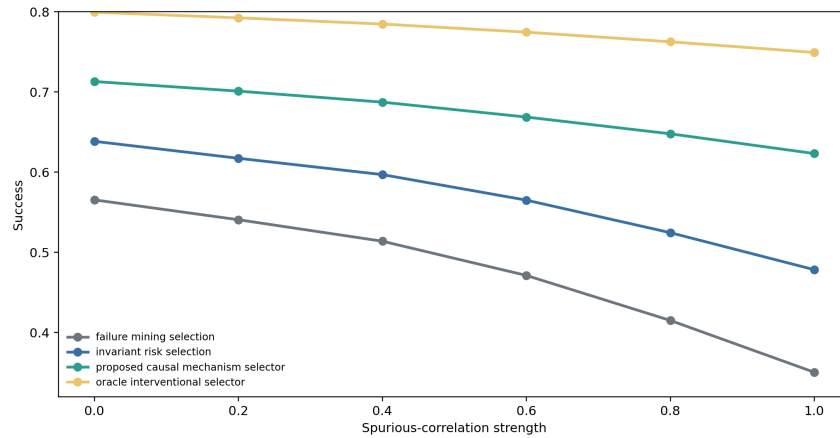


Figure 3: Stress sweep over spurious-correlation strength.

7 LIMITATIONS AND SUBMISSION DECISION

The evidence is local and generated by the released benchmark. It is stronger than the archived scaffold but still lacks real robot experiments, external high-fidelity simulator transfer, selected dataset release, trained policy checkpoints, and hardware videos. The honest terminal state is therefore **STRONG_REVISION**: worth rebuilding aggressively, not yet ICLR-main ready.

REFERENCES

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